SIM1003 Calculus II

Chapter 7 Transcendental Functions

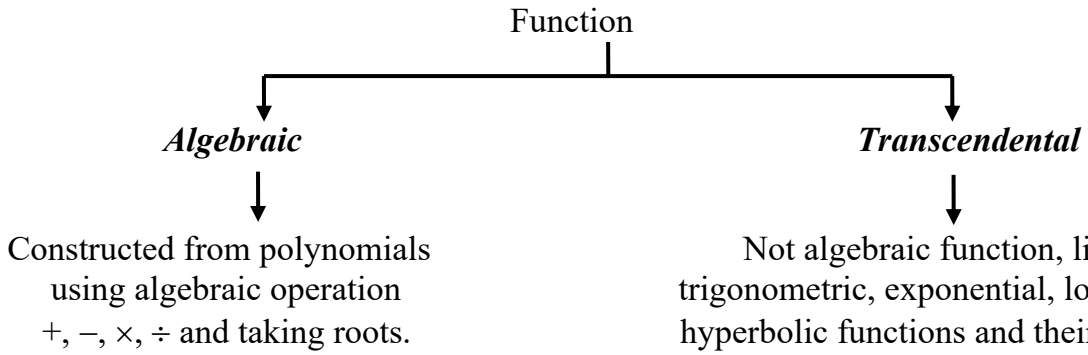
Function

A ***function*** from a set D to a set Y is a rule that assigns a unique element in Y to each element in D.

Polynomial

A ***polynomial*** is an expression of the form $P(x)=a_nx^n+a_{n-1}x^{n-1}+\cdots+a_1x+a_0$ where n is a nonnegative integer and a_0, a_1, …, a_n are constants.

Algebraic function and transcendental function



Example:

$f(x)=x(x^5+1)^3$, $g(x)=\dfrac{x-1}{x+1}$,

$h(x)=x+\sqrt{x+1}$, $\boldsymbol{k(x)=\dfrac{1-\sqrt[3]{x}}{(x^4+x^2-1)^2}}$.

Example:

$f(x)=\sin(x^3+1)$, $g(x)=\ln\left(\dfrac{x-1}{x+1}\right)$,

$h(x)=e^{x+1}$, $p(x)=\cosh\sqrt{x}$,

$q(x)=\sin^{-1}(x^2-1)$, $r(x)=\cosh^{-1}\left(\dfrac{1}{x}\right)$.

7.1 Inverse Functions and their derivatives (Revisit)

Definition 7.1: One-to-one Function
A function $f(x)$ is **one-to-one** on a domain D if $f(x_1)\neq f(x_2)$ whenever $x_1\neq x_2$ in D.

To show *f* is 1-1

Method 1: Use the Horizontal line test.

The horizontal line test for 1-1 function
f is 1-1 $\Leftrightarrow$ its graph intersects each **horizontal line** at most once.

Method 2: By definition.
Show that, $\forall\ x_1, x_2 \in D$, $x_1\neq x_2 \Rightarrow f(x_1)\neq f(x_2)$. Or, equivalently, $f(x_1)=f(x_2)\Rightarrow x_1=x_2$.

Example 7.1.1: Show that $f(x) = x^5 + 2$ is 1-1.

Solution:

$\forall\ x_1, x_2 \in (-\infty, \infty),\quad f(x_1) = f(x_2) \Rightarrow x_1^5 + 2 = x_2^5 + 2 \Rightarrow x_1^5 = x_2^5 \Rightarrow x_1 = \left(x_2^5\right)^{1/5} = x_2.$

$\therefore$ f is 1-1.

How show *f* is not 1-1?

Method: Give **2 values $x_1, x_2 \in D$** such that $f(x_1) = f(x_2)$ but $x_1 \neq x_2$.

Example 7.1.2: Is $f(x) = 1 + \cos x$ 1-1?

Solution:

$D_f = \mathrm{R}$.

$\exists\ 0, 2\pi \in D_f$ such that $f(0) = 1 + 1 = f(2\pi)$ but $0 \neq 2\pi$. $\therefore$ f is not 1-1.

Definition 7.2: Inverse Functions
Suppose that f is 1-1 function on a domain D with range R. The ***inverse function* f^{-1}** is defined by **$f^{-1}(a) = b$ if $f(b) = a$.**
The domain of $f^{-1} = R$, and the range of $f^{-1} = D$.

Remark:

1. f^{-1} is the symbol for ***f inverse* (*inverse of f*)**. **f^{-1} <u>does not mean</u>** $\frac{1}{f}$.

2. **Composing a function and its inverse:**
   $(f^{-1} \circ f)(x) = x\ \ \forall\ x \in D$ and $(f \circ f^{-1})(y) = y\ \ \forall\ y \in D_{f^{-1}}$.

   Example 7.1.3:

   Show that the inverse of $f(x) = 5x$ is $g(x) = \frac{1}{5}x$.

   Solution:

   $g(f(x)) = g(5x) = \frac{1}{5}(5x) = x$ and $f(g(x)) = f\left(\frac{1}{5}x\right) = 5\left(\frac{1}{5}x\right) = x$. $\therefore$ $g = f^{-1}$.

3. No all functions have inverses. **$f^{-1}(x)$ not necessarily exists.**

4. **Only 1-1 *f* have an inverse.**

5. If **f^{-1}** exists, it is **<u>unique</u>**.

6. **$f(f^{-1}(x)) = f^{-1}(f(x)) = x$.**

7. $D_{f^{-1}} = R_f$ and $R_{f^{-1}} = D_f$.

Example 7.1.4: Let $f(x) = x^2 - 4$. Does $f^{-1}(0)$ exist?

Solution:
Suppose $f^{-1}(0) = a$. Then, $f(a) = 0 \Rightarrow a^2 - 4 = 0 \Rightarrow a = \pm 2$.
Since a is not unique, then $f^{-1}(0)$ does not exist.

Alternative method:

$f(2) = 0 = f(-2)$ but $2 \neq -2$, then f is not 1-1 at $x = 0$. $\therefore$ $f^{-1}(0)$ does not exist.

Example 7.1.5: Find the inverse of $f(x) = \sqrt{4x-5}$, if exists.

Solution:
The function is 1-1 in its domain. So, its inverse exists.

$$y = \sqrt{4x-5} \Rightarrow y^2 = 4x - 5 \Rightarrow x = \frac{1}{4}(y^2 + 5) \Rightarrow f^{-1}(x) = \frac{1}{4}(x^2 + 5).$$

$\therefore$ $f^{-1}(x) = \frac{1}{4}(x^2 + 5)$, $x \geq 0$.

Derivatives of Inverse

If $f'(a) \neq 0$, then the slope of $y = f^{-1}(x)$ at (b, a) is given by $\frac{1}{f'(a)}$.

That is $\left.\frac{d}{dx} f^{-1}\right]_{x=b} = \frac{1}{f'(a)} \Rightarrow (f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$ ($\because$ $f(a) = b$

$\Rightarrow$ $\left.\frac{df^{-1}}{dx}\right]_{x=b} = \frac{1}{\left.\frac{df}{dx}\right]_{x=f^{-1}(b)}}$ $\Rightarrow a = f^{-1}(b)$.)

Theorem 7.1: The Derivative Rule for Inverses

If f has an interval I as domain and $f'(x)$ exists and is never zero on I, then f^{-1} is differentiable at every point in its domain.
The value of $(f^{-1})'$ at a point b in the domain of f^{-1} at $a = f^{-1}(b)$ is

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))} \quad \text{or} \quad \left.\frac{df^{-1}}{dx}\right]_{x=b} = \frac{1}{\left.\frac{df}{dx}\right]_{x=f^{-1}(b)}}.$$

Example 7.1.6:

Let $f(x) = x^3 - 2$. Find the value $\frac{df^{-1}}{dx}$ at $x = f(2)$ without finding a formula for $f^{-1}(x)$.

Solution: $\left.\frac{df}{dx}\right]_{x=2} = 3x^2\Big]_{x=2} = 12$. $\therefore$ $\left.\frac{df^{-1}}{dx}\right]_{x=f(2)} = \frac{1}{\left.\frac{df}{dx}\right]_{x=2}} = \frac{1}{12}$.

7.6 Inverse trigonometric functions

All the six basic trigonometric functions are not 1-1. We can **restrict their domains** to intervals on which they are **1-1**, so that each of the functions has an inverse.

Domain restrictions that make the trigonometric functions 1-1

Function	**Restricted Domain = $R_{inverse}$**	**$D_{inverse}$ = Range of the function**
sin x	$\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$	$[-1, 1]$
cos x	$[0, \pi]$	$[-1, 1]$
tan x	$\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$	$(-\infty,\infty)$
cot x	$(0, \pi)$	$(-\infty,\infty)$
sec x	$\left[0,\frac{\pi}{2}\right)\cup\left(\frac{\pi}{2},\pi\right]$	$(-\infty,-1] \cup [1, \infty)$
csc x	$\left[-\frac{\pi}{2},0\right)\cup\left(0,\frac{\pi}{2}\right]$	$(-\infty,-1] \cup [1, \infty)$

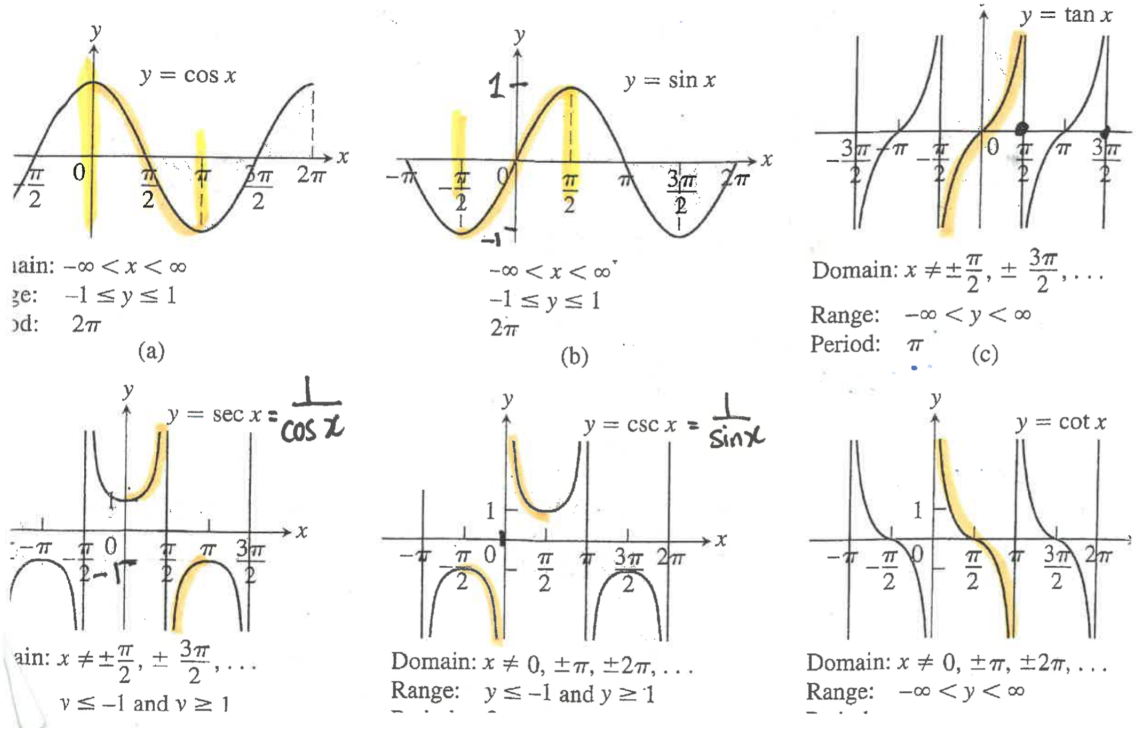
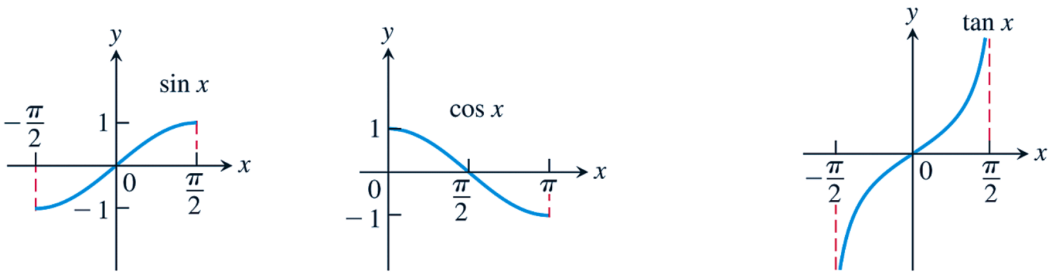


Figure 7.21 Graphs of six basic trigonometric functions



$y = \sin x$
Domain: $[-\pi/2, \pi/2]$
Range: $[-1, 1]$

$y = \cos x$
Domain: $[0, \pi]$
Range: $[-1, 1]$

$y = \tan x$
Domain: $(-\pi/2, \pi/2)$
Range: $(-\infty, \infty)$

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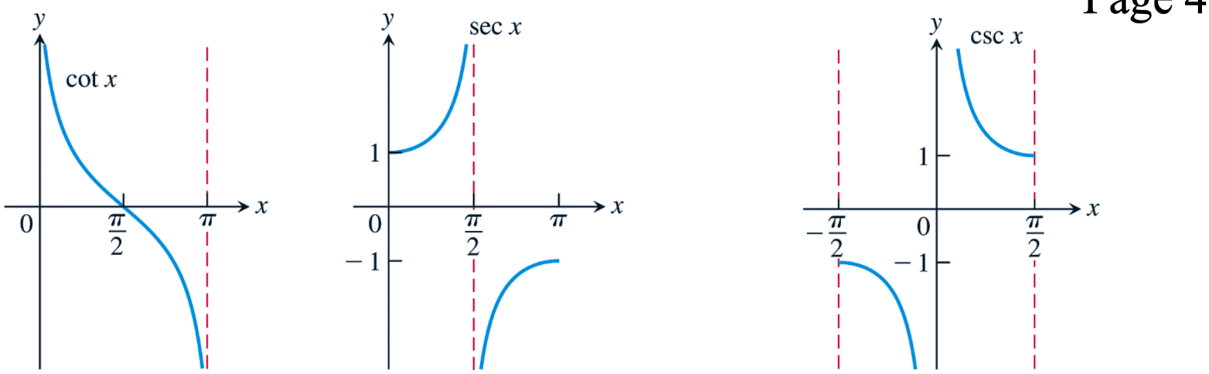


$y = \cot x$
Domain: $(0, \pi)$
Range: $(-\infty, \infty)$

$y = \sec x$
Domain: $[0, \pi/2) \cup (\pi/2, \pi]$
Range: $(-\infty, -1] \cup [1, \infty)$

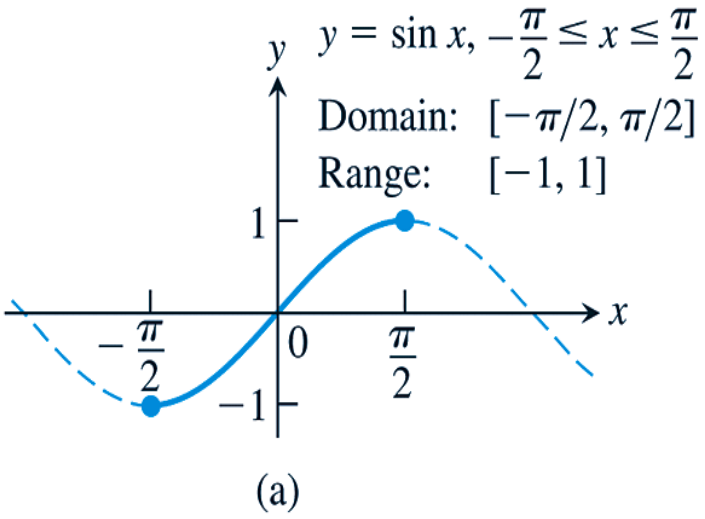
$y = \csc x$
Domain: $[-\pi/2, 0) \cup (0, \pi/2]$
Range: $(-\infty, -1] \cup [1, \infty)$

Figure 7.22 Graphs of six basic trigonometric functions with restricted domains

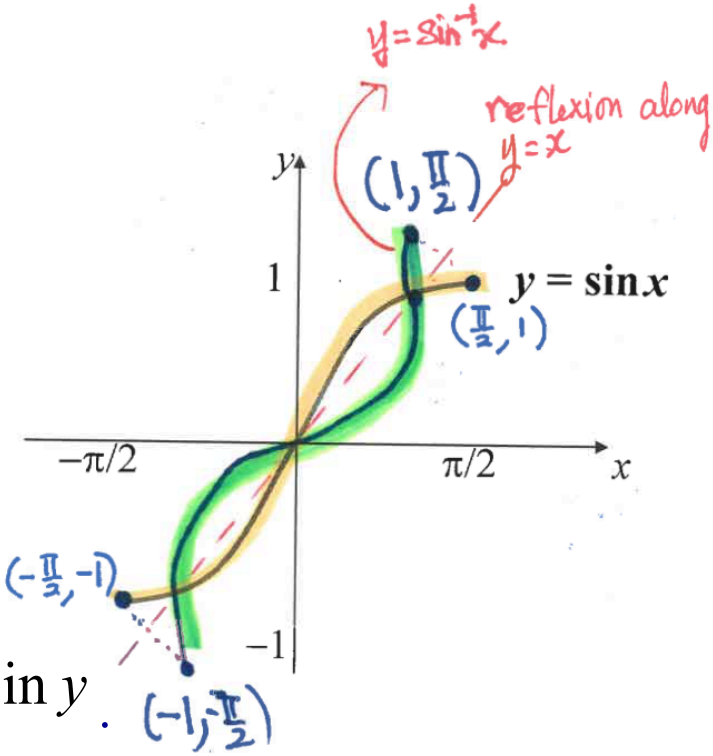
Definition: Arcsine, arccosine, arctangent, Arccotangent, Arcsecant and Arccosecant

1. $y = \sin^{-1} x$ or $y = \arcsin x$ is the angle in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ for which $\sin y = x$.
2. $y = \cos^{-1} x$ or $y = \arccos x$ is the angle in $[0, \pi]$ for which $\cos y = x$.
3. $y = \tan^{-1} x$ or $y = \arctan x$ is the angle in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ for which $\tan y = x$.
4. $y = \cot^{-1} x$ or $y = \operatorname{arccot} x$ is the angle in $(0, \pi)$ for which $\cot y = x$.
5. $y = \sec^{-1} x$ or $= \operatorname{arcsec} x$ is the angle in $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$ for which $\sec y = x$.
6. $y = \csc^{-1} x$ or $= \operatorname{arccsc} x$ is the angle in $\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$ for which $\csc y = x$.

Caution: The "–1" in the definition means "inverse". It does *not* mean reciprocal. For example, the reciprocal of $\sin x$ is $y = (\sin x)^{-1} = \dfrac{1}{\sin x} = \csc x$.



(a)



The graph of $y = \sin^{-1} x$ or $y = \arcsin x$ is obtained by reflection across the line $y = x$. It is a portion of the curve $x = \sin y$.

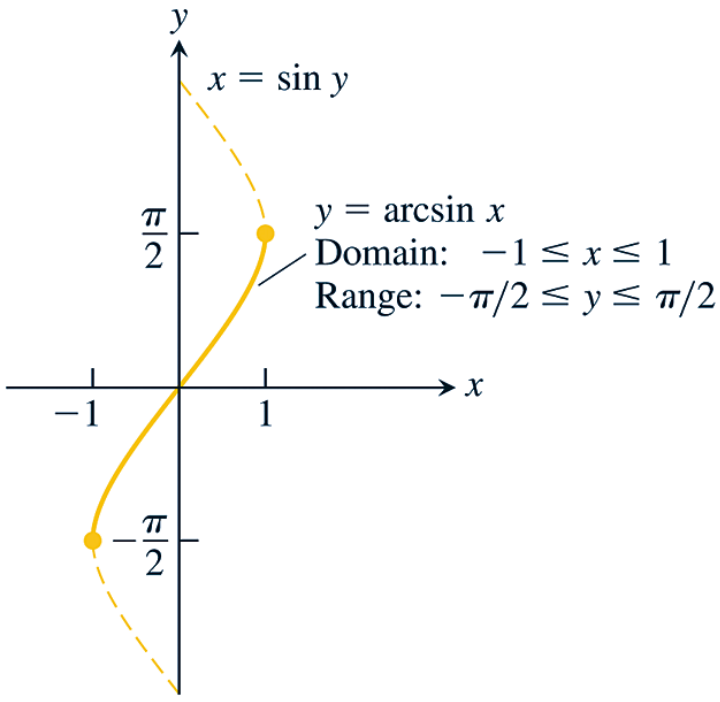
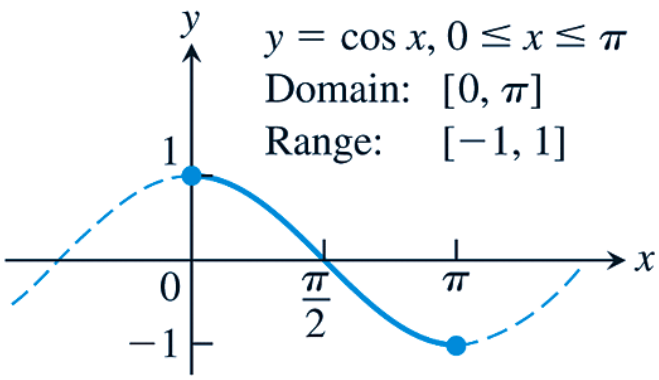
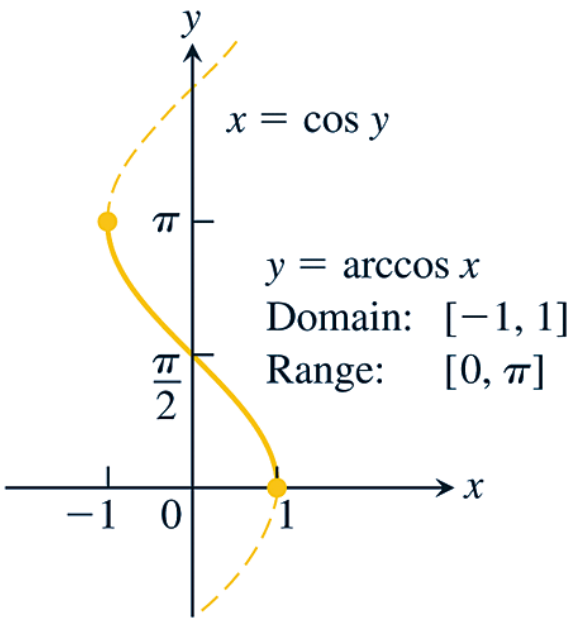


Figure 7.22



(a)

The graph of $y = \cos^{-1} x$ or $y = \arccos x$ is obtained by reflection across the line $y = x$.
It is a portion of the curve $x = \cos y$.



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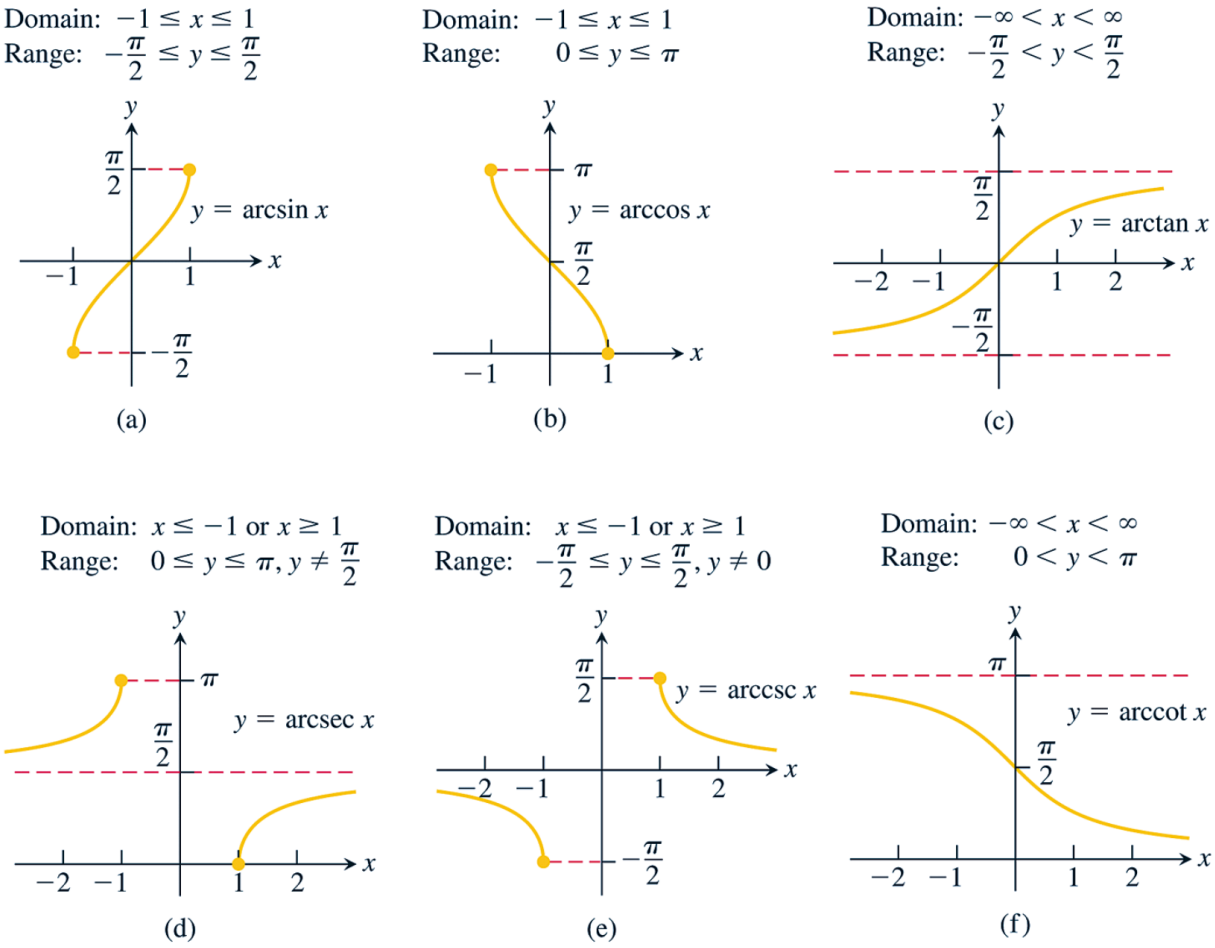
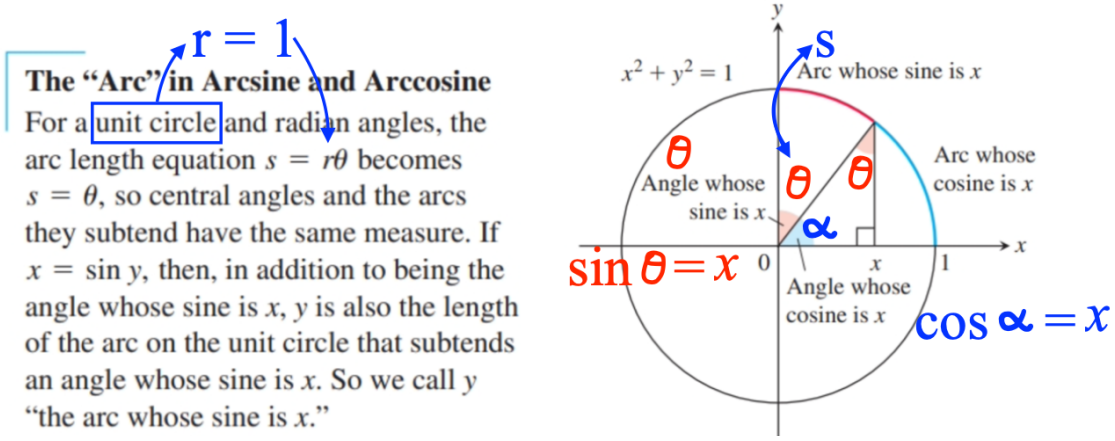


Figure 7.23 Graphs of six basic inverse trigonometric functions

Remarks:

1. The graphs of $y = \sin^{-1} x$ and $y = \tan^{-1} x$ (See Figure7.23)are symmetric about the origin, hence both of the functions are odd functions.

2. The graph of $y = \cos^{-1} x$ is neither symmetric about the y-axis nor the origin, hence it is neither even nor odd.

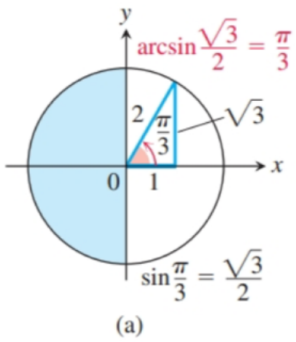


Example 1:

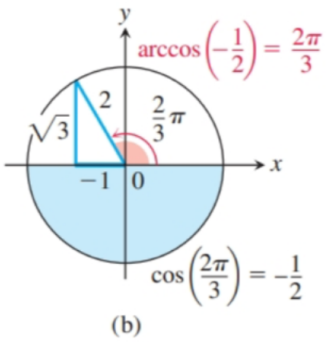
Evaluate (a) $\arcsin\left(\frac{\sqrt{3}}{2}\right)$ (b) $\arccos\left(-\frac{1}{2}\right)$

Using the same procedure illustrated in Example 7.6.1, we can create the following table of common values for arcsine and arccosine functions.

x	$\arcsin x$	$\arccos x$
$\sqrt{3}/2$	$\pi/3$	$\pi/6$
$\sqrt{2}/2$	$\pi/4$	$\pi/4$
$1/2$	$\pi/6$	$\pi/3$
$-1/2$	$-\pi/6$	$2\pi/3$
$-\sqrt{2}/2$	$-\pi/4$	$3\pi/4$
$-\sqrt{3}/2$	$-\pi/3$	$5\pi/6$

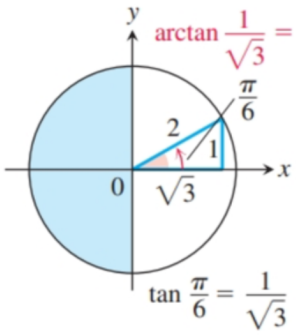


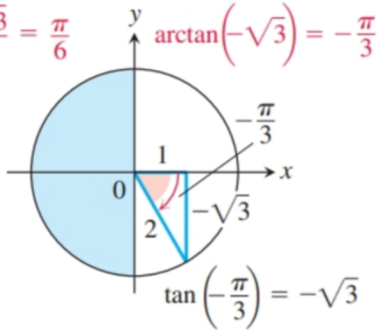
(a)



(b)

The following figures show two values of arctan x.





x	$\arctan x$
$\sqrt{3}$	$\pi/3$
1	$\pi/4$
$\sqrt{3}/3$	$\pi/6$
$-\sqrt{3}/3$	$-\pi/6$
-1	$-\pi/4$
$-\sqrt{3}$	$-\pi/3$

The angles come from the first and fourth quadrants because the range of $\arctan x$ is $(-\pi/2, \pi/2)$. ■

Identities Involving Inverse Trigonometric:

(1) $\sin^{-1}(-x) = -\sin^{-1} x$, $x \in [-1, 1]$. (Arcsine is odd)

(2) $\tan^{-1}(-x) = -\tan^{-1} x$, $x \in (-\infty,\infty)$. (Arctangent is odd)

(3) $\cos^{-1} x + \cos^{-1}(-x) = \pi$, $x \in [-1, 1]$.

(4) $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, $x \in [-1, 1]$.

(5) $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$, $x \in (-\infty,\infty)$.

(6) $\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}$, $x \in (-\infty,-1] \cup [1, \infty)$.

(7) $\sec^{-1} x = \cos^{-1} \frac{1}{x}$, $x \in (-\infty,-1] \cup [1, \infty)$.

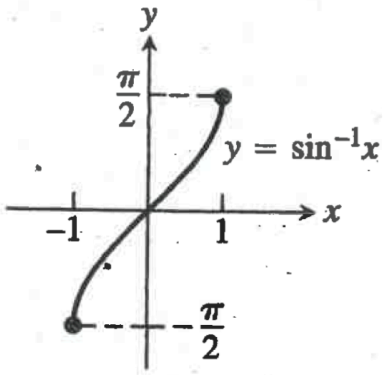
(8) $\csc^{-1} x = \sin^{-1} \frac{1}{x}$, $x \in (-\infty,-1] \cup [1, \infty)$.

(9) $\cot^{-1} x = \tan^{-1} \frac{1}{x}$, $x \in (-\infty, 0) \cup (0, \infty)$.

Example 2:

$\sec^{-1} 2 = \cos^{-1} \frac{1}{2} = \frac{\pi}{3}$. Take note that $\sec^{-1} 2 \neq \frac{1}{\cos^{-1} 2}$ and $\cos^{-1} 2$ is undefined.

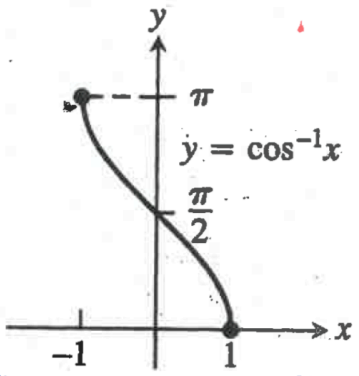
Domain: $-1 \le x \le 1$
Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$



(a) symmetric about the origin
⇒ odd function
⇒ $\sin^{-1}(-x) = -\sin^{-1}x$

Domain: $x \le -1$ or $x \ge 1$
Range: $0 \le y \le \pi, y \ne \frac{\pi}{2}$

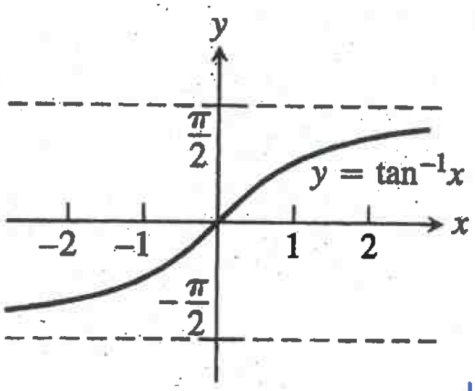
Domain: $-1 \le x \le 1$
Range: $0 \le y \le \pi$



(b) Neither odd nor even

Domain: $x \le -1$ or $x \ge 1$
Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}, y \ne 0$

Domain: $-\infty < x < \infty$
Range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$



(c) symmetric about the origin
⇒ odd ⇒ $\tan^{-1}(-x) = -\tan^{-1}x$

Domain: $-\infty < x < \infty$
Range: $0 < y < \pi$

Remarks:

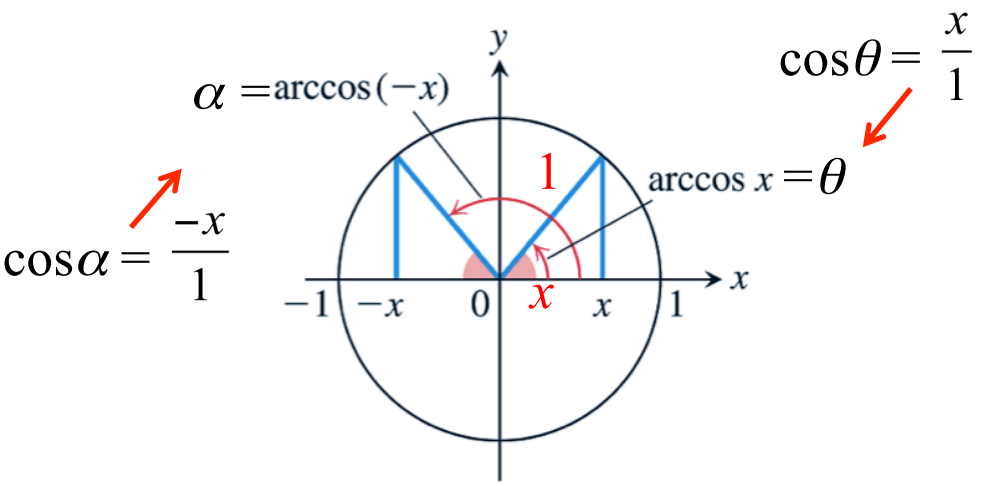
1. The graphs of $y = \sin^{-1} x$ and $y = \tan^{-1} x$ (See Fig. 7.23) are symmetric about the origin, hence both of the functions are odd functions.

2. The graph of $y = \cos^{-1} x$ is neither symmetric about the y-axis nor the origin, hence it is neither even nor odd.

Identities Involving Arcsine and Arccosine

Figure 7.27

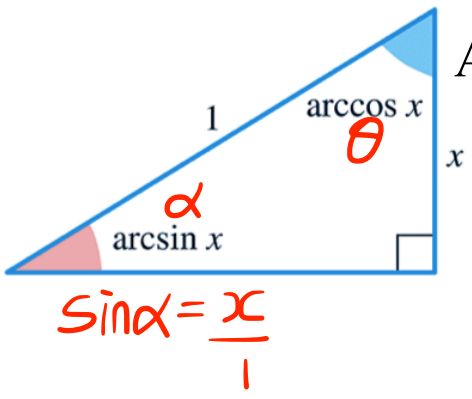
$\arccos x$ and $\arccos(-x)$ are supplementary angles (so their sum is π).



$$\text{Arccos}\,(x) + \text{Arccos}\,(-x) = \pi \qquad (2)$$

Figure 7.28

$\arcsin x$ and $\arccos x$ are complementary angles (so their sum is $\frac{\pi}{2}$).



$$\text{Arcsin}(x) + \text{Arccos}(x) = \frac{\pi}{2} \qquad (4)$$

α + θ

Example 3: $\sec x = \dfrac{1}{\cos x}$, $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots.$

Is $\sec^{-1} x = \dfrac{1}{\cos^{-1} x}$?

Answer: NO. $\sec^{-1} x \neq \dfrac{1}{\cos^{-1} x}$.

$$\sec^{-1} x = \cos^{-1} \frac{1}{x}, \quad x \in (-\infty,-1] \cup [1, \infty).$$

Let $\theta = \sec^{-1} x, \quad \theta \in \left[0 , \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2} , \pi\right]$

Then $\sec\theta = \sec(\sec^{-1} x)$

$\Rightarrow \quad \sec\theta = x$

$\Rightarrow \quad \dfrac{1}{\cos\theta} = x$

$\Rightarrow \quad \cos\theta = \dfrac{1}{x}$

$\Rightarrow \quad \cos^{-1}(\cos\theta) = \cos^{-1}\dfrac{1}{x}$

$\Rightarrow \quad \theta = \cos^{-1}\dfrac{1}{x} \quad \Rightarrow \quad \sec^{-1} x = \cos^{-1}\dfrac{1}{x}.$

The graph of $y = \arctan x$ is symmetric about the origin because it is a branch of the graph $x = \tan y$ that is symmetric about the origin (Figure 7.23c). Algebraically this means that

$$\arctan(-x) = -\arctan x;$$

the arctangent is an odd function. The graph of $y = \operatorname{arccot} x$ has no such symmetry (Figure 7.23d). Notice from Figure 7.23a that the graph of the arctangent function has two horizontal asymptotes: one at $y = \pi/2$ and the other at $y = -\pi/2$.

The inverses of the restricted forms of $\sec x$ and $\csc x$ are chosen to be the functions graphed in Figures 7.23e and 7.24f.

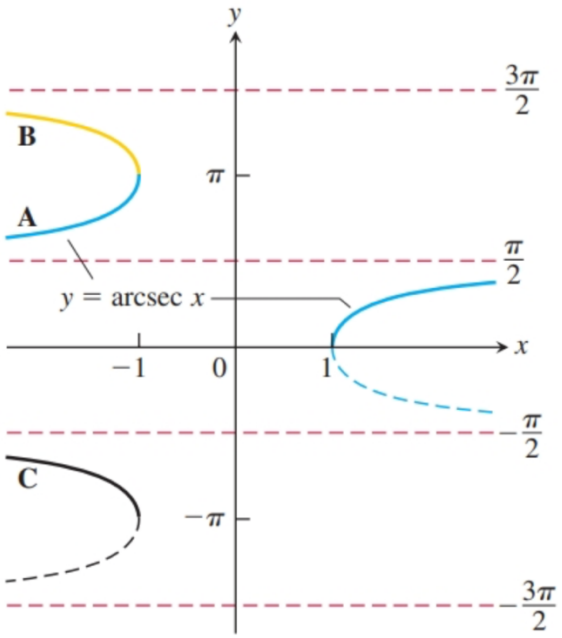


FIGURE 7.29 There are several logical choices for the left-hand branch of $y = \operatorname{arcsec} x$. With choice **A**, $\operatorname{arcsec} x = \arccos(1/x)$, a useful identity employed by many calculators.

Caution There is no general agreement about how to define $\operatorname{arcsec} x$ for negative values of x. We chose angles in the second quadrant between $\pi/2$ and π. This choice makes $\operatorname{arcsec} x = \arccos(1/x)$. It also makes $\operatorname{arcsec} x$ an increasing function on each interval of its domain. Some tables choose $\operatorname{arcsec} x$ to lie in $[-\pi, -\pi/2)$ for $x < 0$ and some texts choose it to lie in $[\pi, 3\pi/2)$ (Figure 7.29). These choices simplify the formula for the derivative (our formula needs absolute value signs) but fail to satisfy the computational equation $\operatorname{arcsec} x = \arccos(1/x)$. From this, we can derive the identity

$$\operatorname{arcsec} x = \arccos\left(\frac{1}{x}\right) = \frac{\pi}{2} - \arcsin\left(\frac{1}{x}\right) \tag{5}$$

by applying Equation (4).

Derivatives of Trigonometric, Inverse Trigonometric, Hyperbolic and Inverse Hyperbolic functions

Table 1

Derivatives of Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

Table 7.7

Derivatives of Hyperbolic Functions

$$\frac{d}{dx}(\sinh x) = \cosh x$$

$$\frac{d}{dx}(\cosh x) = \sinh x$$

$$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

$$\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$$

$$\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$$

$$\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$$

Table 7.3

Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, \quad |x| < 1$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}, \quad |x| < 1$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}, \quad |x| > 1$$

$$\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{|x|\sqrt{x^2-1}}, \quad |x| > 1$$

Table 7.10

Derivatives of Inverse Hyperbolic Functions

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

$$\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}, \quad x > 1$$

$$\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}, \quad |x| < 1$$

$$\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}, \quad |x| > 1$$

$$\frac{d}{dx}(\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1-x^2}}, \quad 0 < x < 1$$

$$\frac{d}{dx}(\operatorname{csch}^{-1} x) = -\frac{1}{|x|\sqrt{1+x^2}}, \quad x \neq 0$$

The Derivative of $y = \arcsin u$

We know that the function $x = \sin y$ is differentiable in the interval $-\pi/2 < y < \pi/2$ and that its derivative, the cosine, is positive there. Theorem 1 in Section 7.1 therefore assures us that the inverse function $y = \arcsin x$ is differentiable throughout the interval $-1 < x < 1$. We cannot expect it to be differentiable at $x = 1$ or $x = -1$ because the tangents to the graph are vertical at these points (see Figure 7.30).

We find the derivative of $y = \arcsin x$ by applying Theorem 1 with $f(x) = \sin x$ and $f^{-1}(x) = \arcsin x$:

Derivative of inverse

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

Theorem 1

Derivative of arcsin x

$$= \frac{1}{\cos(\arcsin x)}$$

$f(u) = \sin u$, $u \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$f'(u) = \cos u$

Change cos u to sin u by using $\cos^2 u + \sin^2 u = 1$

$$= \frac{1}{+\sqrt{1 - \sin^2(\arcsin x)}}$$

$\cos u = +\sqrt{1 - \sin^2 u}$

cos u is positive only because u in 1st & 4th Quadrants

no minus

$[\sin(\arcsin x)]^2$

$$= \frac{1}{\sqrt{1 - x^2}}.$$

$\sin(\arcsin x) = x$

$\cos u = \pm\sqrt{1 - \sin u}$

If u is a differentiable function of x with $|u| < 1$, we apply the Chain Rule to get the general formula

$$\frac{d}{dx}(\arcsin u) = \frac{1}{\sqrt{1 - u^2}}\frac{du}{dx}, \qquad |u| < 1.$$

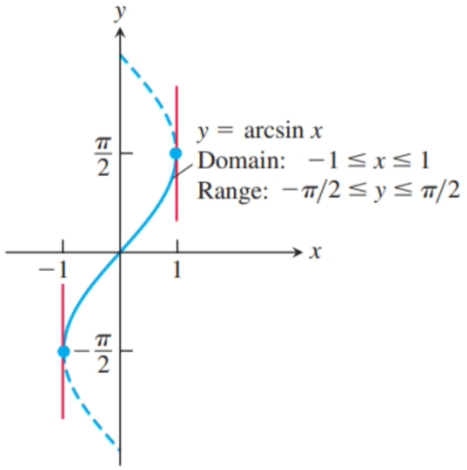


FIGURE 7.30 The graph of $y = \arcsin x$ has vertical tangents at $x = -1$ and $x = 1$.

OR Alternative method

Use implicit differentiation

$$y = \sin^{-1} x \quad , \quad |x| < 1$$

$$\Rightarrow \boxed{\sin y = x} \quad , \quad |y| < \frac{\pi}{2}$$

$$\Rightarrow \frac{d}{dx} \sin y = \frac{dx}{dx}$$

$$\Rightarrow \cos y \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}$$

→ change to x and change $\cos y$ to $\sin y$ using $\sin^2 y + \cos^2 y = 1$

$$= \frac{1}{\sqrt{1 - \sin^2 y}}$$

$\cos y > 0$

$\left(\because -\frac{\pi}{2} < y < \pi/2 \right)$

1st and 4th quadrants

$$= \frac{1}{\sqrt{1 - x^2}}$$

EXAMPLE 4 Using the Chain Rule, we calculate the derivative

$$\frac{d}{dx}(\arcsin x^2) = \frac{1}{\sqrt{1-(x^2)^2}} \cdot \frac{d}{dx}(x^2) = \frac{2x}{\sqrt{1-x^4}}.$$

The Derivative of $y = \arctan u$

We find the derivative of $y = \arctan x$ by applying Theorem 1 with $f(x) = \tan x$ and $f^{-1}(x) = \arctan x$. Theorem 1 can be applied because the derivative of $\tan x$ is positive for $-\pi/2 < x < \pi/2$:

Derivative of arctan x

Change sec u to tan u by using $\sec^2 u = 1 + \tan^2 u$

$$\begin{aligned}
(f^{-1})'(x) &= \frac{1}{f'(f^{-1}(x))} && \text{Theorem 1} \\
&= \frac{1}{\sec^2(\arctan x)} && f'(u) = \sec^2 u \\
&= \frac{1}{1 + \tan^2(\arctan x)} && \sec^2 u = 1 + \tan^2 u \\
&= \frac{1}{1+x^2}. && \tan(\arctan x) = x
\end{aligned}$$

The derivative is defined for all real numbers. If u is a differentiable function of x, we get the Chain Rule form:

$$\frac{d}{dx}(\arctan u) = \frac{1}{1+u^2}\frac{du}{dx}.$$

The Derivative of $y = \text{arcsec } u$

Since the derivative of sec x is positive for $0 < x < \pi/2$ and $\pi/2 < x < \pi$, Theorem 1 says that the inverse function $y = \text{arcsec } x$ is differentiable. Instead of applying the formula in Theorem 1 directly, we find the derivative of $y = \text{arcsec } x$, $|x| > 1$, using implicit differentiation and the Chain Rule as follows:

$$y = \text{arcsec } x$$

$$\sec y = x \qquad \text{Inverse function relationship}$$

$$\frac{d}{dx}(\sec y) = \frac{d}{dx}x \qquad \text{Differentiate both sides.}$$

$$\sec y \tan y \frac{dy}{dx} = 1 \qquad \text{Chain Rule}$$

$$\frac{dy}{dx} = \frac{1}{\sec y \tan y}. \qquad \text{Since } |x| > 1, y \text{ lies in } (0, \pi/2) \cup (\pi/2, \pi) \text{ and } \sec y \tan y \neq 0.$$

y = arcsec x

1st & 2nd Quadrants

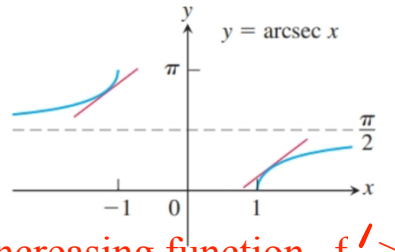
To express the result in terms of x, we use the relationships

Change to sec y

$$\sec y = x \quad \text{and} \quad \tan y = \pm\sqrt{\sec^2 y - 1} = \pm\sqrt{x^2 - 1}$$

to get

$$\frac{dy}{dx} = \pm\frac{1}{x\sqrt{x^2 - 1}}.$$



Increasing function, $f' > 0$

FIGURE 7.31 The slope of the curve $y = \text{arcsec } x$ is positive for both $x < -1$ and $x > 1$.

Can we do anything about the $\pm$ sign? A glance at Figure 7.31 shows that the slope of the graph $y = \text{arcsec } x$ is always positive. Thus,

$\frac{d}{dx}$ arscec x > 0

$$\frac{d}{dx}\text{arcsec } x = \begin{cases} +\dfrac{1}{x\sqrt{x^2 - 1}} & \text{if } x > 1 \\ -\dfrac{1}{x\sqrt{x^2 - 1}} & \text{if } x < -1. \end{cases}$$

Domain of Arcsec x = $(-\infty, -1] \cup [1, \infty)$

|x| > 1

With the absolute value symbol, we can write a single expression that eliminates the "$\pm$" ambiguity:

$$\frac{d}{dx}\text{arcsec } x = \frac{1}{|x|\sqrt{x^2 - 1}}.$$

> 0

If u is a differentiable function of x with $|u| > 1$, we have the formula

$$\frac{d}{dx}(\text{arcsec } u) = \frac{1}{|u|\sqrt{u^2 - 1}}\frac{du}{dx}, \qquad |u| > 1.$$

Figure 7.29

There are several logical choices for the left-hand branch of $y = \operatorname{arcsec} x$. With choice **A**,

$$\operatorname{arcsec} x = \arccos\left(\frac{1}{x}\right),$$

a useful identity employed by many calculators.

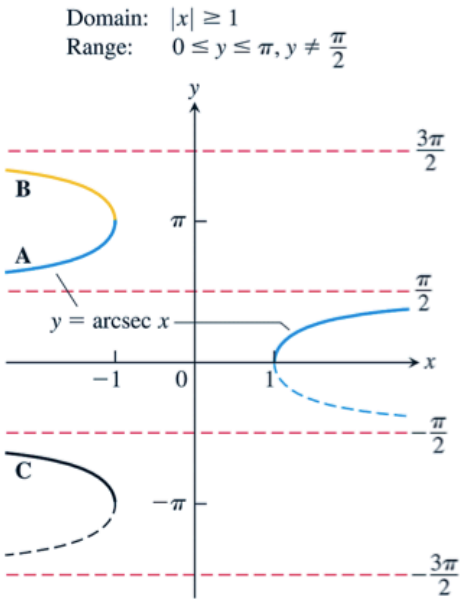
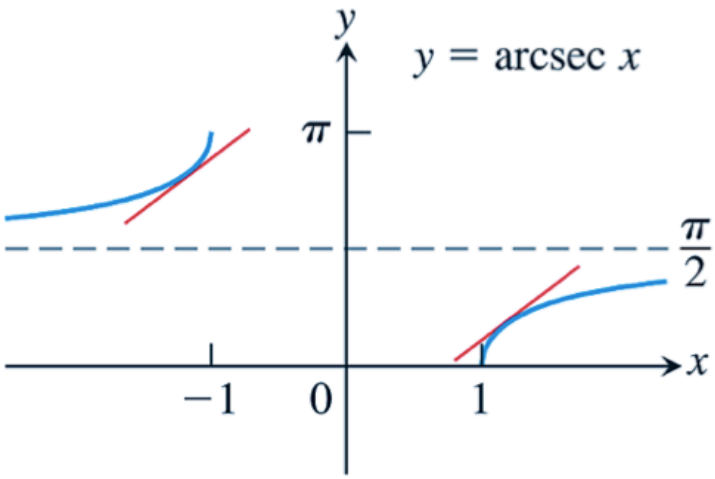


Figure 7.31

The slope of the curve $y = \operatorname{arcsec} x$ is always increasing => f' > 0 is positive for both $x < -1$ and $x > 1$.

|x| > 1 <=> x < -1 or x > 1



EXAMPLE 5 Using the Chain Rule and derivative of the arcsecant function, we find

$$\begin{aligned}\frac{d}{dx}\operatorname{arcsec}(5x^4) &= \frac{1}{|5x^4|\sqrt{(5x^4)^2-1}}\frac{d}{dx}(5x^4)\\ &= \frac{1}{5x^4\sqrt{25x^8-1}}(20x^3) && 5x^4 > 1 > 0\\ &= \frac{4}{x\sqrt{25x^8-1}}.\end{aligned}$$

■

Derivatives of the Other Three Inverse Trigonometric Functions

We could use the same techniques to find the derivatives of the other three inverse trigonometric functions—arccosine, arccotangent, and arccosecant—but there is an easier way, thanks to the following identities.

Inverse Function–Inverse Cofunction Identities

$$\begin{aligned}\arccos x &= \pi/2 - \arcsin x\\ \operatorname{arccot} x &= \pi/2 - \arctan x\\ \operatorname{arccsc} x &= \pi/2 - \operatorname{arcsec} x\end{aligned}$$

We saw the first of these identities in Equation (4). The others are derived in a similar way. It follows easily that the derivatives of the inverse cofunctions are the negatives of the derivatives of the corresponding inverse functions. For example, the derivative of $\cos^{-1} x$ is calculated as follows:

$$\begin{aligned}\frac{d}{dx}(\arccos x) &= \frac{d}{dx}\left(\frac{\pi}{2} - \arcsin x\right) && \text{Identity}\\ &= -\frac{d}{dx}(\arcsin x)\\ &= -\frac{1}{\sqrt{1-x^2}}. && \text{Derivative of arcsine}\end{aligned}$$

The derivatives of the inverse trigonometric functions are summarized in Table 7.3.

TABLE 7.3 Derivatives of the inverse trigonometric functions

1. $\dfrac{d(\arcsin u)}{dx} = \dfrac{1}{\sqrt{1-u^2}}\dfrac{du}{dx}, \quad |u| < 1$

2. $\dfrac{d(\arccos u)}{dx} = -\dfrac{1}{\sqrt{1-u^2}}\dfrac{du}{dx}, \quad |u| < 1$

3. $\dfrac{d(\arctan u)}{dx} = \dfrac{1}{1+u^2}\dfrac{du}{dx}$

4. $\dfrac{d(\operatorname{arccot} u)}{dx} = -\dfrac{1}{1+u^2}\dfrac{du}{dx}$

5. $\dfrac{d(\operatorname{arcsec} u)}{dx} = \dfrac{1}{|u|\sqrt{u^2-1}}\dfrac{du}{dx}, \quad |u| > 1$

6. $\dfrac{d(\operatorname{arccsc} u)}{dx} = -\dfrac{1}{|u|\sqrt{u^2-1}}\dfrac{du}{dx}, \quad |u| > 1$

Integration Formulas

The derivative formulas in Table 7.3 yield three useful integration formulas in Table 7.4. The formulas are readily verified by differentiating the functions on the right-hand sides. Since two notations are regularly used to represent the inverse sine function, we state these formulas using both notations, $\sin^{-1} x$ as well as $\arcsin x$, and similarly for the other inverse trigonometric functions.

TABLE 7.4 Integrals evaluated with inverse trigonometric functions

The following formulas hold for any constant $a > 0$.

1. $\displaystyle\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C$ (Valid for $u^2 < a^2$)

2. $\displaystyle\int \frac{du}{a^2+u^2} = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C$ (Valid for all u)

3. $\displaystyle\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{u}{a}\right| + C$ (Valid for $|u| > a > 0$)

Remarks:
Integrals involve inverse trigonometric functions will be discussed in Chapter 8.

Remark:

We may use the derivatives of $\sin^{-1} x$, $\tan^{-1} x$ and $\sec^{-1} x$ to obtain the derivatives of $\cos^{-1} x$, $\cot^{-1} x$ and $\csc^{-1} x$, respectively.

Example 6: $\dfrac{d}{dx}(\cos^{-1} x) = \dfrac{d}{dx}\left(\dfrac{\pi}{2} - \sin^{-1} x\right) = -\dfrac{1}{\sqrt{1-x^2}}$;

$$\frac{d}{dx}(\cot^{-1} x) = \frac{d}{dx}\left(\frac{\pi}{2} - \tan^{-1} x\right) = -\frac{1}{1+x^2}.$$

Example 7:

Find (1) $\dfrac{d}{dx}\sin^{-1}\sqrt{1-x}$, $0 < x < 1$ (2) $\dfrac{d}{dx}\sec^{-1} e^{2x}$, $x > \dfrac{1}{2}$

(3) $\dfrac{d}{dx}\cos^{-1}\left(\dfrac{x-1}{x+1}\right)$, $x \geq 0$ (4) $\dfrac{d}{dx}\tan^{-1}(1+x^2)$

(5) $\dfrac{d}{dx}\csc^{-1}\left(x^3\right)$ (6) $\dfrac{d}{dx}\sec^{-1}\left(\dfrac{1}{x^2}\right)$, $0 < x < 1$.

Solution:

(1) $$\frac{d}{dx}\sin^{-1}\sqrt{1-x} = \frac{1}{\sqrt{1-\left(\sqrt{1-x}\right)^2}}\frac{d}{dx}\sqrt{1-x}$$

$$= \frac{1}{\sqrt{x}}\left(\frac{1}{2}\right)(1-x)^{-1/2}(-1) = -\frac{1}{2\sqrt{x(1-x)}}.$$

(2) $$\frac{d}{dx}\sec^{-1} e^{2x} = \frac{1}{\left|e^{2x}\right|\sqrt{\left(e^{2x}\right)^2 - 1}}\frac{d}{dx}e^{2x} = \frac{2e^{2x}}{e^{2x}\sqrt{e^{4x}-1}}$$

$$= \frac{2}{\sqrt{e^{4x}-1}} \quad \left(\because e^{2x} > 0 \quad \forall x \quad \Rightarrow \quad |e^{2x}| = e^{2x}\right).$$

(3) $$\frac{d}{dx}\cos^{-1}\left(\frac{x-1}{x+1}\right) = \frac{d}{dx}\left(\frac{\pi}{2} - \sin^{-1}\left(\frac{x-1}{x+1}\right)\right) = \frac{-1}{\sqrt{1-\left(\frac{x-1}{x+1}\right)^2}}\frac{d}{dx}\left(\frac{x-1}{x+1}\right)$$

$$= -\frac{|x+1|}{\sqrt{4x}}\left(\frac{2}{(x+1)^2}\right) = -\frac{|x+1|}{\sqrt{4x}}\left(\frac{2}{(x+1)^2}\right)$$

$$= \frac{-1}{\sqrt{x}(x+1)} \quad \left(\because x \geq 0 \quad \Rightarrow \quad |x+1| = x+1\right).$$

(4) $\dfrac{d}{dx}\tan^{-1}(1+x^2) = \dfrac{1}{1+(1+x^2)^2}\dfrac{d}{dx}(1+x^2) = \dfrac{2x}{2+2x^2+x^4}.$

(5) $\dfrac{d}{dx}\csc^{-1}(x^3) = \dfrac{d}{dx}\left(\dfrac{\pi}{2} - \sec^{-1}(x^3)\right)$

$$= -\frac{1}{|x^3|\sqrt{(x^3)^2-1}}\frac{d}{dx}(x^3) = -\frac{3x^2}{|x^3|\sqrt{x^6-1}}.$$

(6)

Example:

Show that $\sin^{-1}\left(\dfrac{x-1}{x+1}\right) = 2\tan^{-1}\sqrt{x} - \dfrac{\pi}{2}$ for $x \geq 0$.

Solution:

$$\frac{d}{dx}\sin^{-1}\left(\frac{x-1}{x+1}\right) = \frac{1}{\sqrt{1-\left(\dfrac{x-1}{x+1}\right)^2}}\frac{d}{dx}\left(\frac{x-1}{x+1}\right) = \frac{1}{\sqrt{\dfrac{(x+1)^2-(x-1)^2}{(x+1)^2}}}\left(\frac{(x+1)-(x-1)}{(x+1)^2}\right)$$

$$= \frac{|x+1|}{\sqrt{4x}}\left(\frac{2}{(x+1)^2}\right) = \frac{1}{\sqrt{x}(x+1)} \qquad \left(\because x>0 \Rightarrow |x+1| = x+1\right).$$

$$\frac{d}{dx}2\tan^{-1}\sqrt{x} = \frac{2}{1+(\sqrt{x})^2}\frac{d}{dx}\sqrt{x} = \frac{2}{1+x}\left(\frac{1}{2\sqrt{x}}\right)$$

$$= \frac{1}{\sqrt{x}(x+1)} = \frac{d}{dx}\sin^{-1}\left(\frac{x-1}{x+1}\right).$$

$\therefore \sin^{-1}\left(\dfrac{x-1}{x+1}\right) = 2\tan^{-1}\sqrt{x} + C$, C is a constant.

Take $x = 0$, $\sin^{-1}(-1) = 2\tan^{-1}(0) + C \Rightarrow -\dfrac{\pi}{2} = 0 + C \Rightarrow C = -\dfrac{\pi}{2}$.

$\therefore \quad \sin^{-1}\left(\dfrac{x-1}{x+1}\right) = 2\tan^{-1}\sqrt{x} - \dfrac{\pi}{2}.$